

SWE-451 · CLC PROJECT MANAGEMENT PRESENTATION

Choosing the right project management model for Congruent Software

A consulting recommendation from The Big 4 Consulting Agency

Alec Huish

Developer

Keny Beauzile

Developer

Jordan Faina

Infrastructure SE

Elijah Burlend

Operational SE

GitHub Projects board: [add your board link here]

AGENDA

What we'll cover today

01

Congruent's business

What the company does and why it needs a better PM model

02

Six models

Waterfall, Agile, Scrum, Kanban, Lean and Hybrid: pros, cons, tools

03

Roles and the SDLC

Balance across dev, infrastructure and ops, plus fit with the life cycle

04

Cost and benefits

Year-one cost to adopt each model and what it pays back

05

Recommendation

The model we believe fits Congruent best, and why

06

Resources and rollout

What our team needs to put the model in place

THE CLIENT

Congruent Software at a glance

1994

Year founded

3,000+

Projects delivered

95%

Customer retention

2

Countries: U.S. and India

A global IT consulting and software development partner for enterprises, SMBs and ISVs.

- Headquarters in Bellevue, Washington
- Offshore delivery center in Chennai, India
- Microsoft and Salesforce partner
- Already uses AI tools such as GitHub Copilot and Jira AI plugins in delivery

Four lines of work, two rhythms



Custom and product development

Custom apps, SaaS products, MVPs, AI and machine learning



Enterprise platform projects

Dynamics 365, Salesforce and SharePoint rollouts



Cloud and DevOps

Azure and AWS migration, CI/CD pipelines, managed cloud



Support and staffing

Ongoing support, QA and testing, IT staff augmentation

Project work

Client contracts with set scope, budget and go-live dates

Continuous work

Support tickets and requests that never stop arriving

Split teams

Bellevue and Chennai are about 12.5 hours apart

OUR METHOD

Six criteria, weighted for Congruent

Criterion	Weight	Why it matters to Congruent
Fits fixed-bid client contracts	25%	Most platform projects are sold with set scope and price
Adapts to changing requirements	20%	Custom and AI work changes as clients learn
Works for distributed teams	15%	Bellevue and Chennai share little working time
Balances dev, infra and ops	15%	All three roles deliver every engagement
Cost to adopt	15%	Training and tools come out of project margin
Builds in continuous improvement	10%	Protects 95% retention as the company grows

Each model is scored 1 (poor) to 5 (excellent) on every criterion. The weighted results appear on slide 16.

Waterfall

Sequential phases (requirements, design, build, test, deploy), each signed off before the next one begins.

Pros

- Scope, cost and dates fixed up front
- Strong documentation for audits
- Matches fixed-bid client contracts

Cons

- Change is slow and expensive
- Testing comes late, so defects do too
- Client sees working software only at the end

Tools gained

- Gantt charts and MS Project
- Phase-gate sign-offs
- Requirements traceability matrix

Continuous improvement: lessons learned only at project close

Role balance: infra and ops engaged mostly at deployment

Agile

A value-driven approach (Agile Manifesto): deliver working software in short iterations and adapt to client feedback.

Pros

- Adapts as client needs change
- Frequent demos build client trust
- Problems surface early

Cons

- Hard to fix a price on open scope
- Needs an engaged, available client
- Less upfront documentation

Tools gained

- Product backlog and user stories
- Azure Boards or GitHub Projects
- CI/CD pipelines

Continuous improvement: a retrospective after every iteration

Role balance: cross-functional teams include all roles from day one

Scrum

Fixed sprints of one to four weeks, three accountabilities (Product Owner, Scrum Master, Developers) and five events.

Pros

- Predictable cadence and velocity
- Clear roles and ownership
- Sprint reviews keep the client involved

Cons

- Fixed sprints clash with urgent tickets
- Role overhead on small projects
- Needs trained Scrum Masters

Tools gained

- Sprint backlog and burndown charts
- Daily scrum, review, retrospective
- PSM and PSPO certification path

Continuous improvement: the sprint retrospective is a required event

Role balance: strong for developers; ops can feel bolted on

Kanban

Visualize work on a board, limit work in progress (WIP), and pull new work only as capacity frees up. No fixed iterations.

Pros

- Ideal for support tickets and requests
- Easy to layer onto today's process
- Exposes bottlenecks quickly

Cons

- No built-in planning cadence or deadlines
- Weak for large fixed-scope builds
- WIP limits need discipline

Tools gained

- Kanban boards and WIP limits
- Cumulative flow diagrams
- Cycle-time and lead-time metrics

Continuous improvement: flow metrics show where to improve

Role balance: excellent for ops and infra queues

Lean Development

Remove anything that does not add value for the client, guided by seven principles adapted from Toyota's production system.

Pros

- Cuts waste: handoffs, waiting, rework
- Builds quality in early
- Faster delivery at lower cost

Cons

- A philosophy, not a step-by-step process
- Needs culture change and buy-in
- "Decide late" can clash with fixed bids

Tools gained

- Value stream mapping
- Waste (muda) analysis
- Automated testing to build quality in

Continuous improvement: kaizen is the core of the model

Role balance: "optimize the whole" unites dev, infra and ops

Waterfall-Agile Hybrid

Waterfall-style planning and gates for scope, budget and contracts; Agile sprints for designing, building and testing.

Pros

- Fixed-bid contracts with agile delivery
- Client gets demos and predictability
- Gates satisfy enterprise governance

Cons

- Two rule sets to learn and manage
- Can slide into "waterfall in disguise"
- Needs clear handoffs between phases

Tools gained

- Phase gates plus sprint boards
- Azure DevOps Boards, Repos, Pipelines
- A Kanban lane for support work

Continuous improvement: sprint retros plus a review at each gate **Role balance:** gates give infra and ops planned checkpoints

BALANCE ACROSS ROLES

How each model serves all three roles

Model	Developers	Infrastructure	Operations	Improvement loop
Waterfall	Low: rework is costly	Good: capacity planned early	Low: engaged at the end	End of project
Agile	High: fast feedback	Medium	Medium	Every iteration
Scrum	High: clear sprint goals	Medium	Low: tickets break sprints	Every sprint
Kanban	Medium	High: queue-based work	High: built for tickets	Continuous metrics
Lean	Medium	High: fewer handoffs	High: less waiting	Kaizen culture
Hybrid	High: sprints	High: planned gates	High: Kanban lane	Retros plus gate reviews

RELATION TO THE SDLC

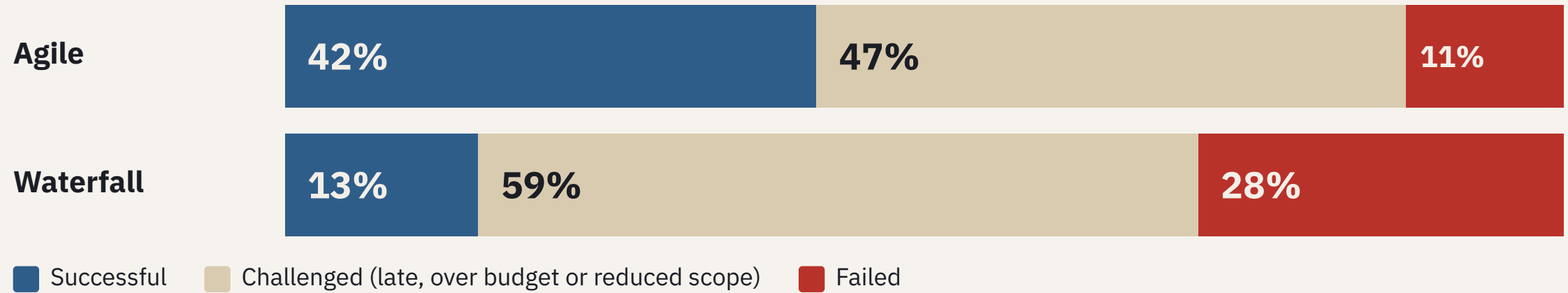
One life cycle, six paths through it



Model	How it moves through the SDLC
Waterfall	Every phase once, in order; no going back without a change request
Agile	Repeats design, build and test in short iterations; release often
Scrum	A full mini-SDLC inside every one-to-four-week sprint
Kanban	Each work item flows through all phases on its own; strongest in maintenance
Lean	Does not replace phases; removes waste and waiting between them
Hybrid	Plan and requirements once, sprints for design to test, gated deploy, Kanban maintenance

INDUSTRY EVIDENCE

Iterative projects succeed about three times as often



Takeaway: iterative delivery more than triples the success rate and cuts outright failure by more than half.

COST ANALYSIS AND BENEFITS

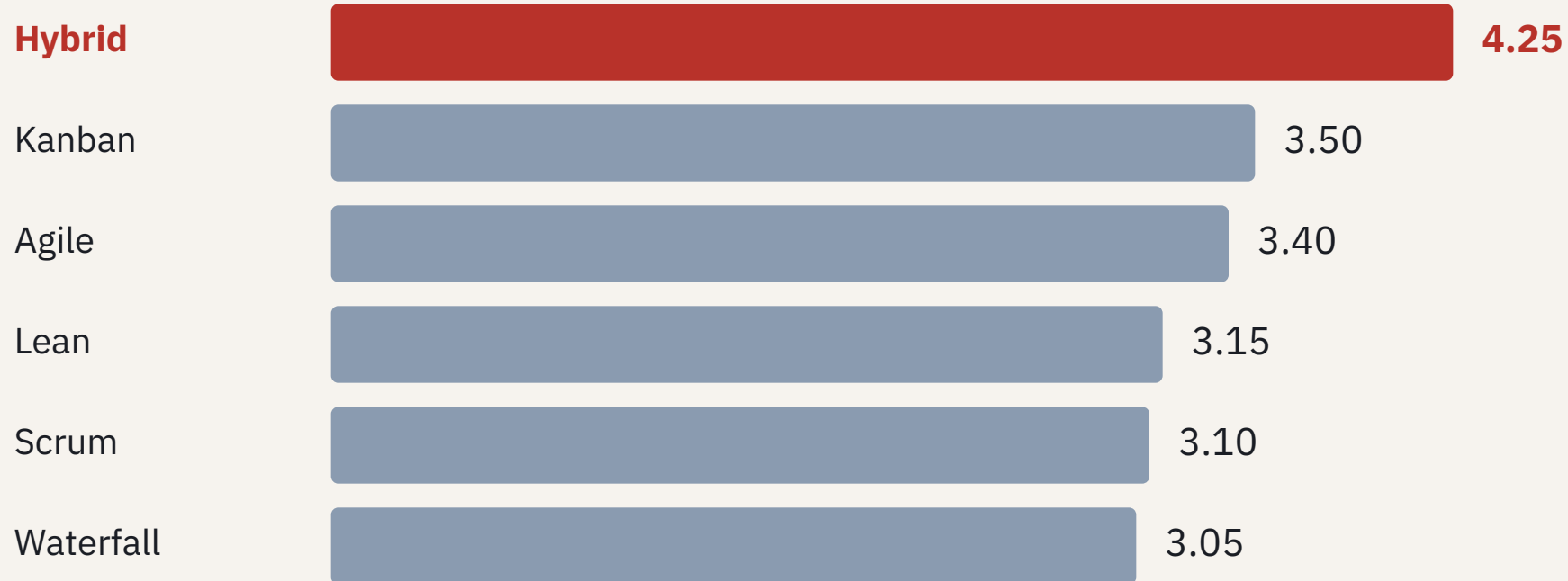
Year-one cost to adopt each model

Model	Training	Certs	Tooling	Year-1 total	Key benefit
Waterfall	\$9,600	\$0	\$2,520	\$12,120	Lowest cost; teams already know it
Kanban	\$19,200	\$0	\$2,520	\$21,720	Faster ticket resolution
Agile	\$38,400	\$0	\$2,520	\$40,920	Fewer failed projects
Lean	\$38,400	\$0	\$2,520	\$40,920	Less rework and waiting
Hybrid	\$48,000	\$1,600	\$2,520	\$52,120	Keeps fixed bids, gains agile delivery
Scrum	\$57,600	\$1,600	\$2,520	\$61,720	Predictable velocity, clear roles

Assumptions (our estimates): 40-person pilot group; \$60/hour blended rate; training of 4 (Waterfall), 8 (Kanban), 16 (Agile, Lean), 20 (Hybrid) or 24 (Scrum) hours per person; eight PSM or PSPO exams at \$200; Azure DevOps Basic, first 5 users free then 35 × \$6 × 12 months.

WEIGHTED COMPARISON

The hybrid scores highest overall



Weighted score out of 5 across the six criteria on slide 5. Only the hybrid scores 4 or higher on fixed-bid fit, adaptability and role balance at once.

OUR RECOMMENDATION

Adopt a Waterfall-Agile hybrid, with Kanban for support work

Sells the way clients buy

Fixed scope and price stay; delivery inside becomes iterative

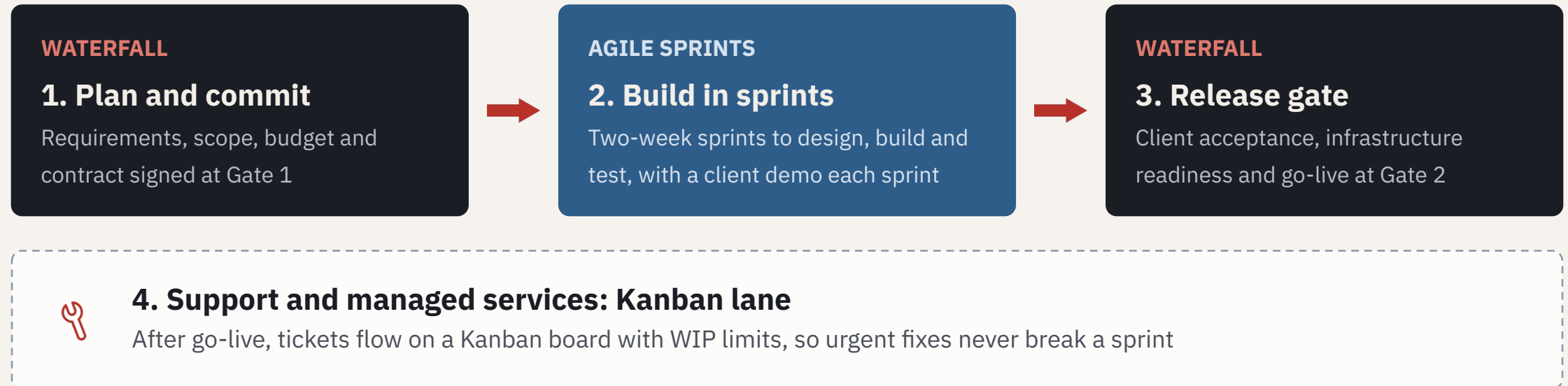
Works for every role

Sprints for developers, gates for infrastructure, Kanban for ops

Best value, not lowest cost

About \$52K in year one for the highest weighted score

One engagement, start to finish



RESOURCES REQUIRED

What our team needs to make it work



People

Our 4 consultants, 4 Scrum Masters, 4 Product Owners, a sponsor



Tools

Azure DevOps Boards, Repos, Pipelines: about \$2,520 a year



Training

20 hours per pilot member and 8 Scrum.org exams (about \$49,600)



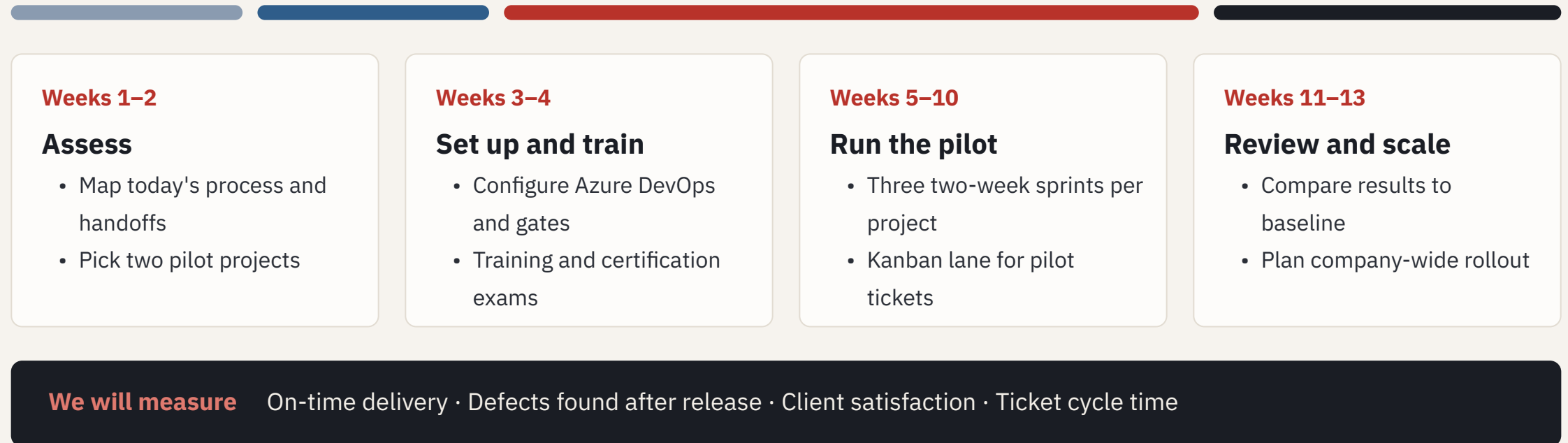
Time

A 13-week pilot on two client projects before company-wide rollout

Consultant	Role in the rollout
Alec and Keny, Developers	Coach dev teams on sprints, backlogs and CI/CD practices
Jordan, Infrastructure SE	Configure Azure DevOps, pipelines, environments and release gates
Eli, Operational SE	Stand up the Kanban support lane, metrics and rollout governance

ROLLOUT PLAN

A 13-week pilot, then scale



SUMMARY

Plan like Waterfall. Build like Agile. Support with Kanban.

A Waterfall-Agile hybrid keeps Congruent's fixed-bid model while delivering iteratively, for about \$52K in year one.

Questions?

References

Agile Genesis. (2023). *Agile vs. waterfall: Comparing success rates in project management*. <https://www.agilegenesis.com>

Anderson, D. J. (2010). *Kanban: Successful evolutionary change for your technology business*. Blue Hole Press.

Beck, K., et al. (2001). *Manifesto for agile software development*. <https://agilemanifesto.org>

Congruent Software. (2026). *Software development services company*. <https://www.congruentsoft.com>

Congruent Software. (n.d.). *About Congruent Software Inc.* <https://www.congruentsoft.com/company-overview.aspx>

Microsoft. (n.d.). *Pricing for Azure DevOps*.

<https://azure.microsoft.com/en-us/pricing/details/devops/azure-devops-services>
Poppendieck, M., & Poppendieck, T. (2003). *Lean software development: An agile toolkit*. Addison-Wesley.

Royce, W. W. (1970). Managing the development of large software systems. *Proceedings of IEEE WESCON*, 1–9.

Schwaber, K., & Sutherland, J. (2020). *The Scrum guide*. <https://scrumguides.org>

Scrum.org. (n.d.). *Professional Scrum Master I certification*. <https://www.scrum.org/assessments/professional-scrum-master-i-certification>